

Dr. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech.

Branch : Petrochemical Engg.

Semester :VI

Subject Code & Name: BTPCC601 REACTION ENGINEERING-II

Max Marks: 60

Date:18/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	To maximize the formation of R in the simultaneous reactions $A + B \rightarrow R \quad r_R = 2C_A^{0.5} C_B^2$ $A + B \rightarrow S \quad r_S = 1.5C_A C_B$	Level 2	1
	a. Low C_A, Low C_B b. Low C_A, High C_B c. High C_A, Low C_B d. none of these		
2	In a consecutive reaction system, $A \xrightarrow{E_1} B \xrightarrow{E_2} C$ when E_1 is greater than E_2, the yield of B increases with the.	Level 1,2	1
	a. Decrease in temperature. b. Increase in initial concentration of A c. Decreases in initial concentration of A d. Increase in temperature		
3	For a series reaction, the relative yield	Level 1,2	1
	a. is always greater for plug-flow reactor than for the single CSTR of the same volume b. statement in (a) is wrong c. decreases with increasing conversion. d. both (a) and (c) hold good		
4	For the uni-molecular type elementary reactions $A \xrightarrow{k_1} B \xrightarrow{k_2} C$, the fractional yield of B in MFR for the given conversion of A	Level 1,2	1
	a. remains constant with increase in k_2/k_1 b. increases with the increase in k_2/k_1 c. decreases with the increase in k_2/k_1 d. none of these		
5	The enzyme which can catalyze the conversion of glucose to ethyl alcohol is	Level 1	1
	a. Invertase b. Maltase c. Diastase d. Zymase		

6	Consider an enzyme-catalyzed fermentation reactions $A \xrightarrow{\text{Enzyme}} T$ With experimental rate $r_R = \frac{k[A][E_0]}{[M]+[A]}$ Given $[M] \gg [A]$ The order of the reaction is ----- a. 0 b. 1 c. 0 or 1 depends on conditions d. can't be determined	Level 1,2	1
7	In solid Catalyzed reactions, the diffusional effects are more likely to affect the overall rate of reaction for a. Fast reactions in catalysts of small pore diameter b. Slow reactions in catalysts of small pore diameter c. Fast reactions in catalysts of large pore diameter d. Slow reactions in catalysts of large pore diameter	Level 1,2	1
8	Knudsen diffusion is directly proportional to a. T b. $\sqrt{T}$ c. $1/\sqrt{T}$ d. T^2	Level 1	1
9	For a packed bed reactor, the presence of a long tail in the residence time distribution curve is an indication of a. Ideal plug flow b. Bypass c. Dead zone d. Channeling	Level 1,2	1
10	For an ideal mixed flow reaction (CSTR) the exit age distribution E (t) is given by a. dirac delta function b. step function c. ramp function d. none of the above	Level 1	1
11	The E-curve for a non-ideal reactor defines the fraction of fluid having age between t and t + dt a. at the inlet b. at the outlet c. in the reactor d. averaged over the inlet and outlet	Level 1	1
12	Pick out the wrong statement. a. In a multistep reaction, the slowest step is the rate determining step. b. In general, the rate of a reaction becomes triple for every 10°C rise in temperature. c. Slow chemical reactions have generally high values of activation energy. d. Molecularity of a reaction cannot be zero but the order of a reaction can be zero.	Level 1,2	1
Q. 2	Solve the following.		12
A)	For the elementary reaction $A \xrightarrow{k_1} B \xrightarrow{k_2} C$ $B \xrightarrow{k_3} D$ Find $C_{R,\max}$ for CSTR	Level 2,3	6
B)	The following liquid-phase reactions were carried out in a CSTR at 325 K.	Level 1, 2	6

	$3A \longrightarrow B + C \quad -r_{1A} = k_{1A} C_A \quad k_{1A} = 7.0 \text{ min}^{-1}$ $2C + A \longrightarrow 3D \quad r_{2D} = k_{2D} C_C^2 C_A \quad k_{2D} = 3.0 \text{ dm}^6/\text{mol}^2 \cdot \text{min}$ $4D + 3C \longrightarrow 3E \quad r_{2E} = k_{3E} C_D C_C \quad k_{3E} = 2.0 \text{ dm}^3/\text{mol} \cdot \text{min}$ The concentrations measured inside the reactor were $C_A = 0.10$, $C_B = 0.93$, $C_C = 0.51$, and $C_D = 0.049$ all in mol/dm^3 . i) What are r_{1A} , r_{2A} and r_{3A} ? ii) What are r_{1B} , r_{2B} and r_{3B} ? iii) What are r_{1C} , r_{2C} and r_{3C} ? iv) What are r_{1D} , r_{2D} and r_{3D} ? v) What are r_{1E} , r_{2E} and r_{3E} ? vi) What are the net rates of formation of A, B, C, D, and E?																														
Q.3	Solve the following.		12																												
A)	Give the informative account of the following (i) Active intermediates (ii) PSSH	Level 1,2	6																												
B)	Using the PSSH, develop a rate law for the formation of HBr for the mechanism below. Initiation: $\text{Br}_2 \xrightarrow{k_1} 2\text{Br}^*$ $\text{Br}^* + \text{H}_2 \xrightarrow{k_2} \text{HBr} + \text{H}^*$ Propagation: $\text{H}^* + \text{Br}_2 \xrightarrow{k_3} \text{HBr} + \text{Br}^*$ $\text{H}^* + \text{HBr} \xrightarrow{k_4} \text{H}_2 + \text{Br}^*$ Termination: $2\text{Br}^* \xrightarrow{k_5} \text{Br}_2$	Level 2,3	6																												
Q.4	Solve Any Two of the following.		12																												
A)	Discuss the significance of Sherwood, Reynolds and Schmidt numbers used in forced convection mass transfer.	Level 2,3	6																												
B)	For mass transfer limited reactions in packed bed show that $\frac{C_A}{C_{A0}} = \exp - \left[\frac{k_c a_c}{U} Z \right]$	Level 2,3	6																												
C)	Discuss diffusion through a stagnant gas and show that $W_A = cD_{AB} \nabla \ln y_B$.	Level 2,3	6																												
Q.5	Solve Any Two of the following.		12																												
A)	Give the brief account for zero parameter models.	Level 1,2	6																												
B)	A sample of tracer hythane at 320 K was injected as a pulse to a reactor, and the effluent concentration was measured as a function of time, resulting in the data shown in table <table border="1" style="margin: 10px auto; border-collapse: collapse;"> <tr> <td>t (min)</td> <td>0</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> <td>7</td> <td>8</td> <td>9</td> <td>10</td> <td>12</td> <td>14</td> </tr> <tr> <td>C (g·m⁻³)</td> <td>0.0</td> <td>1.0</td> <td>5.0</td> <td>8.0</td> <td>10.0</td> <td>8.0</td> <td>6.0</td> <td>4.0</td> <td>3.0</td> <td>2.2</td> <td>1.5</td> <td>0.6</td> <td>0.0</td> </tr> </table> Calculate the mean residence time and the variance for the rector.	t (min)	0	1	2	3	4	5	6	7	8	9	10	12	14	C (g·m ⁻³)	0.0	1.0	5.0	8.0	10.0	8.0	6.0	4.0	3.0	2.2	1.5	0.6	0.0	Level 2,3,4	6
t (min)	0	1	2	3	4	5	6	7	8	9	10	12	14																		
C (g·m ⁻³)	0.0	1.0	5.0	8.0	10.0	8.0	6.0	4.0	3.0	2.2	1.5	0.6	0.0																		
C)	Discuss the different types of boundary conditions used for heterogeneous reaction.	Level 2,3	6																												
Q.6	Solve Any Two of the following.		12																												
A)	Discuss tanks – in – series model versus dispersion model.	Level 1,2	6																												

B)	The liquid-phase, second-order dimerization $2A \longrightarrow B \qquad r_A = -kC_A^2$ For which $k = 0.01 \text{ dm}^3/\text{mol}\cdot\text{min}$ is carried out at a reaction temperature of 320 K. The feed is pure A with $C_{A0} = 8 \text{ mol/ dm}^3$. The reactor is non-ideal and perhaps could be modeled as two CSTRs with interchange. The reactor volume is 1000 dm^3 and the feed rate for our dimerization is going to be $25 \text{ dm}^3/\text{min}$. Following table shows results for a tracer test on this reactor. Find the bounds on the conversion for different possible degrees of micro mixing for the RTD of this reactor. Tracer test on tank reactor: $N_o = 100\text{g}$, $v = 25 \text{ dm}^3/\text{min}$. <table><tr><td>t (min)</td><td>0</td><td>5</td><td>10</td><td>15</td><td>20</td><td>30</td><td>40</td><td>50</td><td>70</td><td>100</td><td>150</td><td>200</td></tr><tr><td>C (mg dm³)</td><td>112</td><td>95.8</td><td>82.2</td><td>70.6</td><td>60.9</td><td>45.6</td><td>34.5</td><td>26.3</td><td>15.7</td><td>7.67</td><td>2.55</td><td>0.90</td></tr></table>	t (min)	0	5	10	15	20	30	40	50	70	100	150	200	C (mg dm ³)	112	95.8	82.2	70.6	60.9	45.6	34.5	26.3	15.7	7.67	2.55	0.90	Level 2,3,4	6
t (min)	0	5	10	15	20	30	40	50	70	100	150	200																	
C (mg dm ³)	112	95.8	82.2	70.6	60.9	45.6	34.5	26.3	15.7	7.67	2.55	0.90																	
C)	Show that RTD for n CSTRs in series, $E(t)$ $E(t) = \frac{t^{n-1}}{(n-1)! \tau_1^n} e^{-t/\tau_1}$	Level 2,3	6																										
*** End ***																													